

Department of Mechanical Engineering

UNIT 3

Wind Energy & Biomass Energy

Programme	B.E (UG)
Semester	I/II
Course Name	Renewable Energy Sources
Course Code	ETC144
Course Credits	3-0-0
Unit 3	Wind Energy: Properties of wind, availability of wind energy in India, wind velocity and power from wind; major problems associated with wind power, Basic components of wind energy conversion system (WECS); Classification of WECS- Horizontal axis- single, double and muliblade system. Vertical axis- Savonius and darrieus types. Biomass Energy: Introduction; Photosynthesis Process; Biofuels; Biomass Resources; Biomass conversion technologies -fixed dome; Urban waste to energy conversion; Biomass gasification (Downdraft).

UNIT 3

Wind Energy & Biomass Energy

Wind energy: Wind is used to produce electricity by converting the kinetic energy of air in motion into electricity. In modern wind turbines, wind rotates the rotor blades, which convert kinetic energy into rotational energy. This rotational energy is transferred by a shaft which to the generator, thereby producing electrical energy.

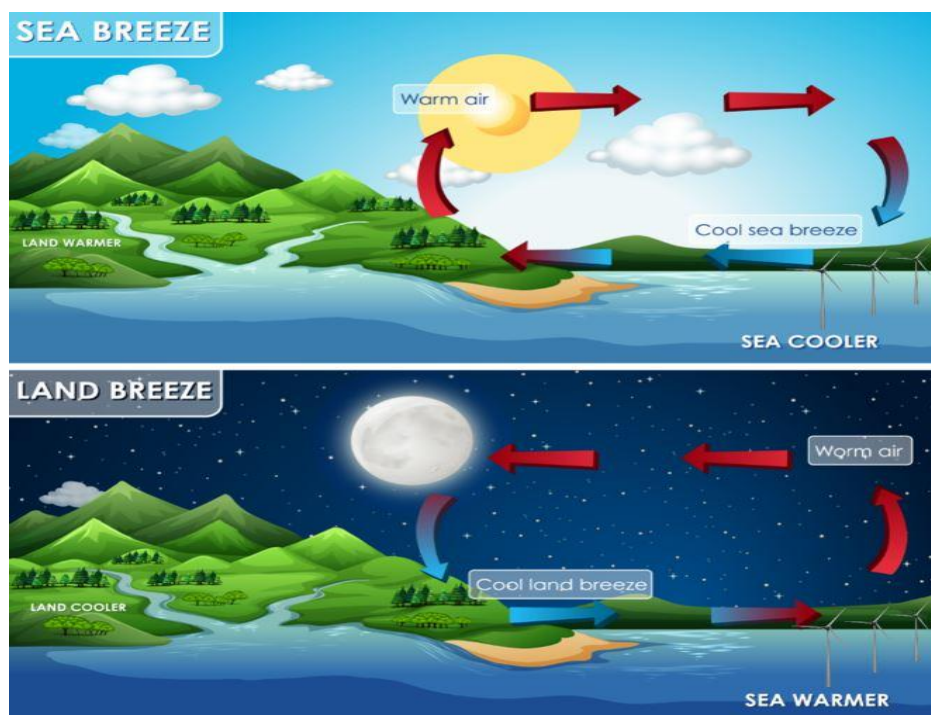
$$\text{Wind energy} = \text{Kinetic energy of air} = \frac{1}{2} mv^2 = \frac{1}{2} \rho Av^3$$

where ρ is density of air, A is swept area of turbine and v is the velocity of the air

Wind is caused by uneven heating of the earth's surface by the sun. Because the earth's surface is made up of different types of land and water, the earth absorbs the sun's heat at different rates. One example of this uneven heating is the daily wind cycle.

The daily wind cycle : During the day, air above land heats up faster than air above water. Warm air above land expands and rises, and heavier, cooler air rushes in to take its place, creating wind. At night, the winds are reversed because air cools more rapidly over land than it does over water.

In the same way, the atmospheric winds that circle the earth are created because the land near the earth's equator is hotter than the land near the North Pole and the South Pole.





Types of wind energy:

1. Onshore wind energy: The wind energy extracted from earth's land surface is called Onshore wind energy.



2. Offshore wind energy: The wind energy extracted from sea surface is called Onshore wind energy.



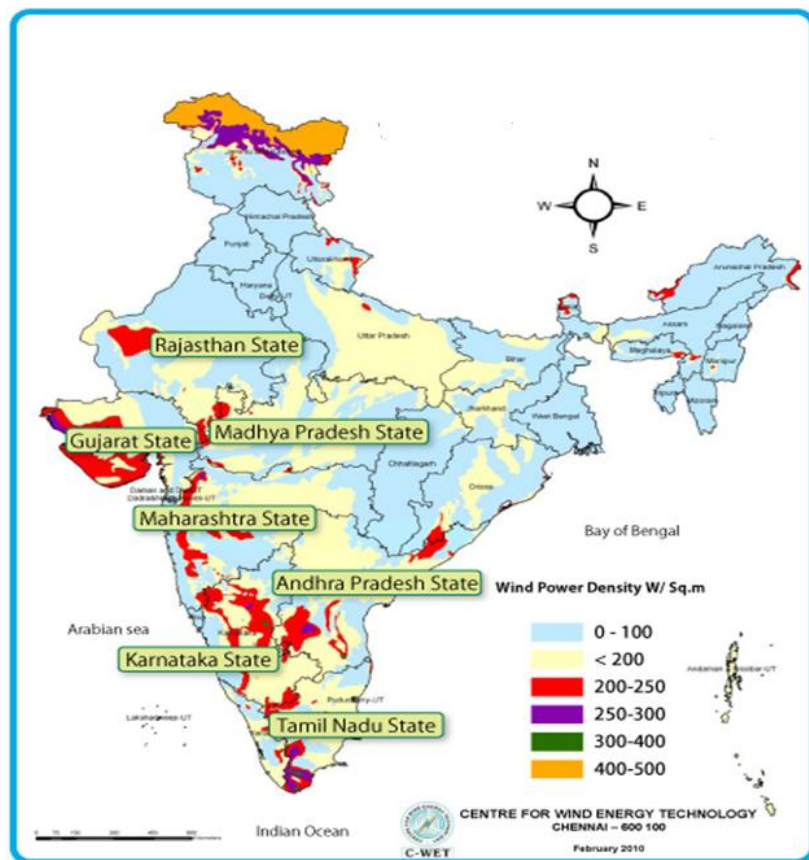
Wind power has grown rapidly since 2000, driven by R&D, supportive policies and falling costs. Global installed wind generation capacity – both onshore and offshore – has increased by a factor of 98 in the past two decades, jumping from 7.5 GW in 1997 to some 733 GW by 2018 according to IRENA's (International Renewable Energy Agency) data. Onshore wind capacity grew from 178 GW in 2010 to 699 GW in 2020, while offshore wind has grown proportionately more, but from a lower base, from 3.1 GW in 2010 to 34.4 GW in 2020. Production of wind power increased by a factor of 5.2 between 2009 and 2019 to reach 1412 TWh.



Properties of wind energy:

- 1) It is intermittent in nature.
- 2) The wind power systems do not contribute any pollution to atmosphere.
- 3) The utilization of wind energy for power generation does not consume any fuel and it is also free from transportation of fuel.
- 4) It is a renewable source of energy.
- 5) It is available in dilute form.
- 6) The availability of wind energy varies over a day and also with seasons. This necessitates the use of storage energy.
- 7) In a small scale, wind energy is cheaper. When produced on a large scale, it is a competitive power generation system.

Availability of wind energy in India: As of 31 December 2022, the total installed wind power capacity was 41.93 Giga-watts (GW), the fourth largest installed wind power capacity in the world. Wind power capacity is mainly spread across the southern, western, and north-western states.



Wind velocity and power from wind

Due to motion of wind, it possesses certain energy. The wind mills are the devices that convert kinetic energy of the wind into mechanical energy. The output from the wind mill depends upon

- the wind speed
- the cross section of the wind swept by rotor and
- the overall conversion efficiency of the rotor, transmission system and generator or pump.

We know that

$$m = \rho AV$$

where

$$m = \text{Mass of air}$$

$$\rho = \text{Air density}$$

$$A = \text{Area through which air is traversing}$$

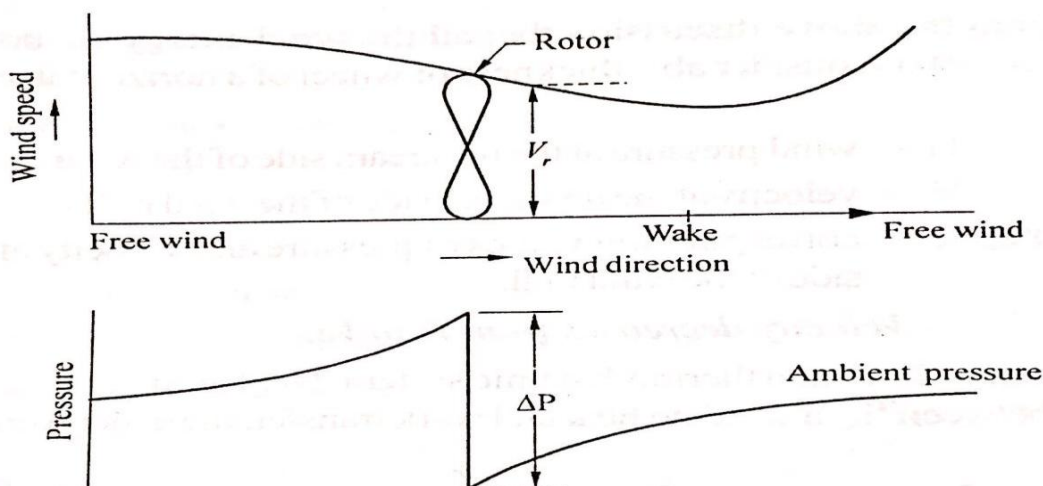
$$V = \text{Wind speed}$$

But kinetic energy, $KE = \frac{1}{2} m V^2$

substitute the value of m in the above equation, we get

$$KE = \frac{1}{2} \rho A V \cdot V^2$$

$$\therefore KE = \frac{1}{2} \rho \cdot A \cdot V^3 \text{ Watts}$$



Variations of wind speed and pressure across wind rotor

$$\text{Power coefficient} = \frac{\text{Power of the wind rotor}}{\text{Power available in the wind}}$$

Maximum Power Coefficient or Betz limit is **0.593**. This means that only 59.3% of kinetic energy of air can be converted into electric energy not more than that.



Advantages

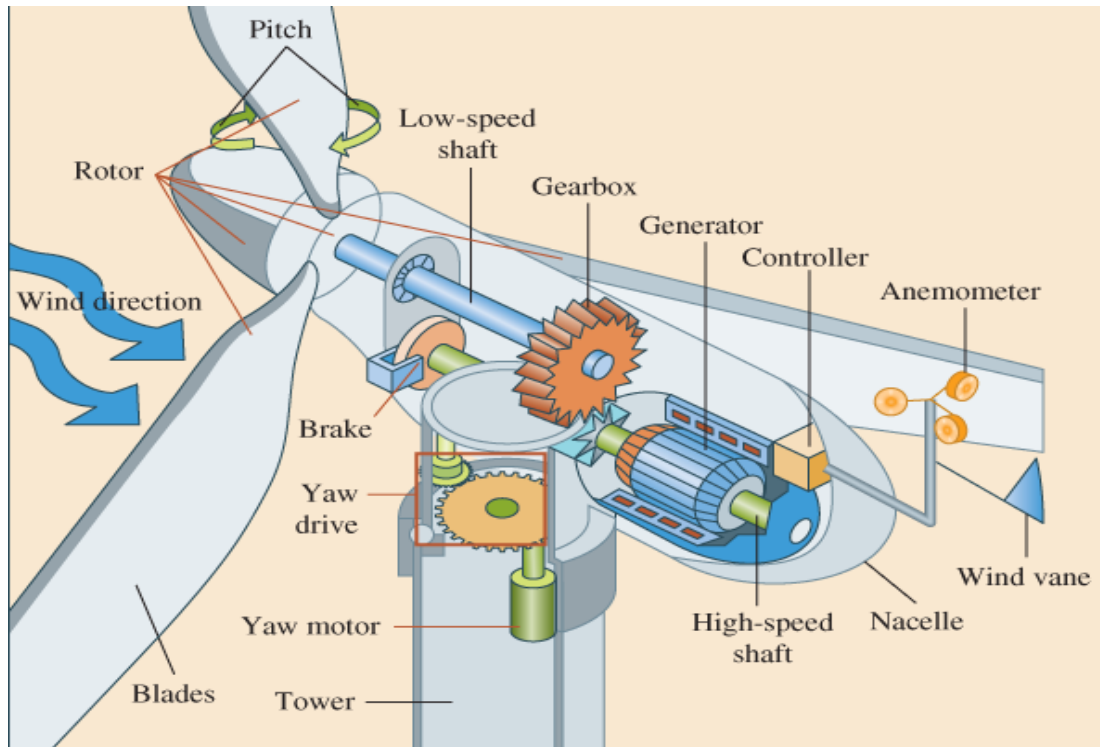
- 1) Wind energy is free inexhaustible and does not need transportation.
- 2) Erection time is less compared to other power plants.
- 3) It is simple with minimum operational expenditures.
- 4) It can be economically constructed at rural areas for power generation, water lifting, etc.
- 5) It does not pollute the environment in any manner as the conventional fossil fuel plants do.
- 6) During monsoon periods, when water level in reservoirs comes down, then wind power generated can be more economical than hydel power.

Major problems associated with wind power

1. It's availability is not continuous ie intermittent. This makes the plant design difficult and it requires a storage system.
2. The power output is very low even with large sized wind mill rotors.
3. The use of special controls and suitable materials, and costly designs are essential to avoid smashing of the wind mill plants.
4. The wind mill has very low power coefficient and can have a maximum value of 0.593.
5. It is not economical to develop wind power on a large scale. The electricity generation, storage and distribution is much costlier.
6. Energy is available in very dilute form and large areas are needed to install wind mills.

A **windmill** is defined as a machine that converts the kinetic energy of the wind into mechanical energy. All the blades of windmill always rotate in a clockwise direction. The first windmill was designed in the year 1854 by Daniel Halladay from the United States.

Components of wind mill or wind turbine



Foundation: For any wind turbine, the foundation gives support to the tower because the wind turbine includes different parts which weigh in tonnes.

Tower: A tower is used to give support to the rotor hub and nacelle on the top of the window turbine. The materials used to make this are concrete, tubular steel, or steel lattice. While designing this turbine, the height of the tower is very important because wind speed enhances with height. So taller towers allow these turbines to capture a huge amount of energy & produce more electricity. Generally, the output power of a wind turbine enhances by increasing its height & also decreasing the turbulence within the wind. There are different wind turbine towers available like tubular, lattice, guyed wind, tilt upwind & free standing.

Wind Turbine Blades: These blades are mainly used to remove the kinetic energy (KE) of wind & change it to mechanical energy. These types of blades are designed with wood-epoxy or fiber-glass-reinforced polyester. These turbines include a minimum of one and maximum multiple blades depending on the design. Most of the horizontal axis wind turbines include three blades that are connected to the rotor hub. In earlier days, multiple blades based turbines are used as a single blade, two-blade and three blades for grinding & pumping water, etc.



Rotor-Drive shaft: The rotor drive shaft is the rotating shaft attached to the rotor wind blades. Which transfer the power from rotor to generator by the help of gear box.

Nacelle: The nacelle includes different components which are used to operate the wind turbine efficiently like the gearbox, brakes, controller, low & high-speed shafts & generator. It is arranged at the top of a tower & a wind vane is arranged on the nacelle.

Anemometer: Anemometer is the device which measures the velocity of the air. Which is attached the rear end of the wind turbine.

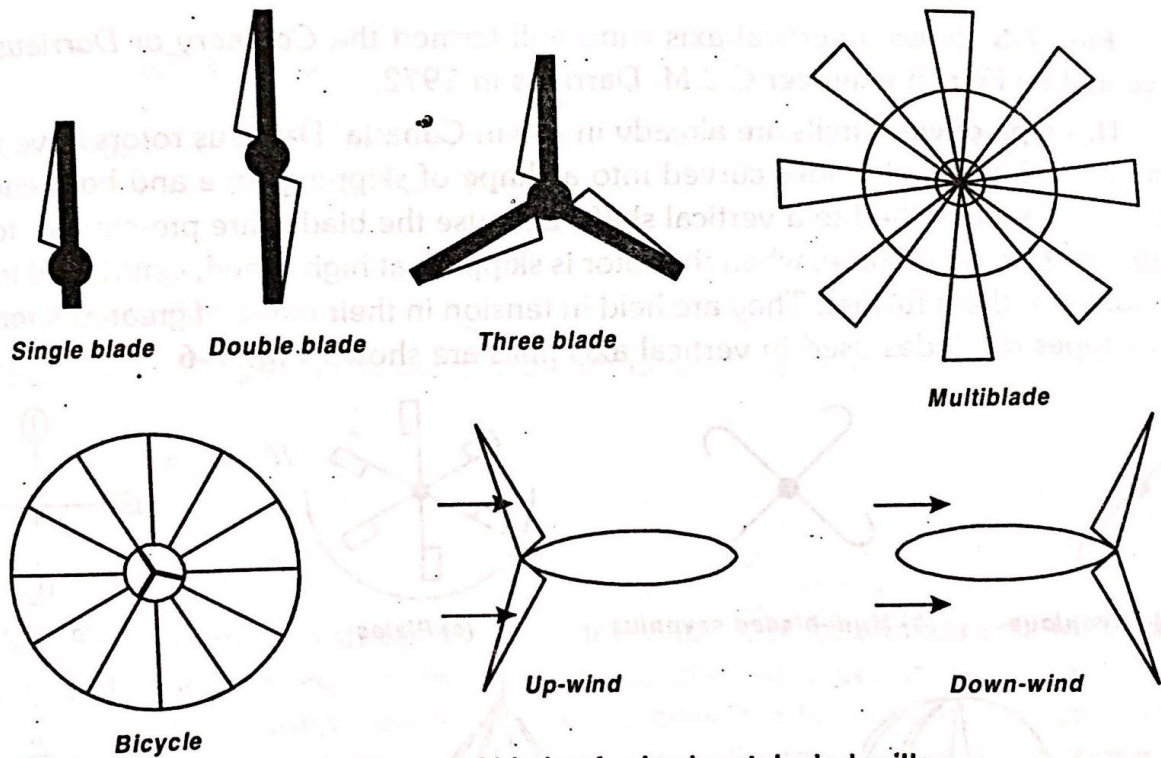
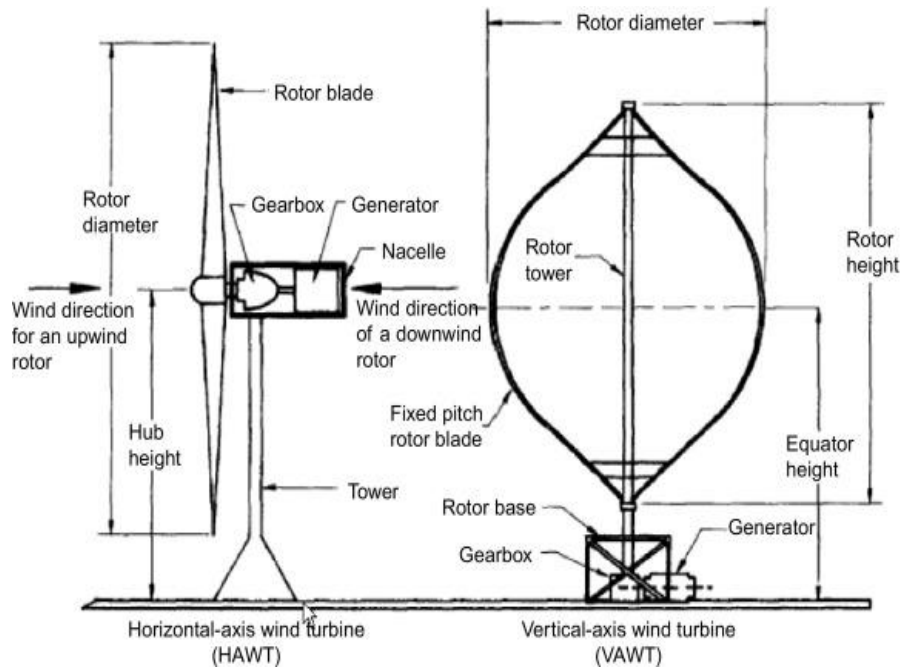
Hub: A rotor hub is used to connect a shaft and rotor blade of the wind turbine. The hub includes blade bearings, bolts, internals & a pitch system. These are designed with cast iron, welded sheet steel & forged steel. These are available in two types like Hinge-less hub & Teetering hub.

Gear Box: In wind turbines, a gearbox is used to change high torque power with low-speed which is received from a rotor blade to low torque power with high speed. This power is used for the generator. The gearbox is connected in between the generator and main shaft for enhancing rotational speeds from 30 – 60 rpm to 1000 – 1800 rpm. Gearboxes are made with different materials like superior quality alloys, aluminum cast iron, stainless steel, etc. In wind turbines, there are three types of gearboxes are used like Planetary, Helical, and Worm.

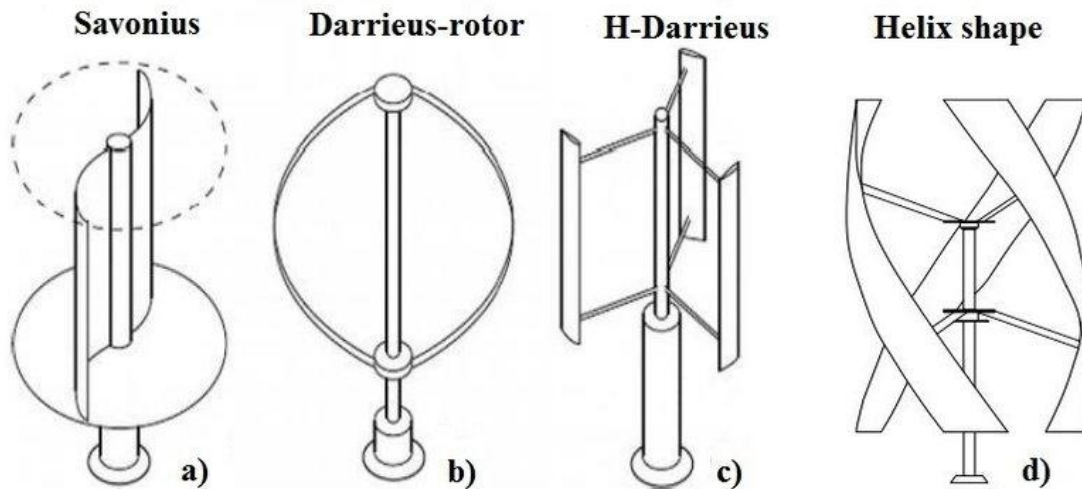
Yaw drive system: The yaw drive is an important component of the horizontal axis wind turbines' yaw system. To ensure the wind turbine is producing the maximal amount of electric energy at all times, the yaw drive is used to keep the rotor facing into the wind as the wind direction changes.

Generator: The rotating mechanical energy of the gearbox is given to the generator through the shaft. It works on 'Faraday's law of electromagnetic induction principle. So it changes the energy from mechanical to electrical.

Classification of wind mill based on axis of rotation



Types of blades for horizontal wind mills



Types of vertical wind mills blades

Site selection consideration

1. High annual average wind speed
2. Availability of anemometry data
3. Availability of wind $V(t)$ curve at the proposed site.
4. Wind structure at the proposal site
5. Altitude of site
6. Land cost
7. Nature of ground
8. Distance to roads and railway



Difference between HAWT and VAWT

Basis of Difference	Horizontal Axis Wind Turbine	Vertical Axis Wind Turbine
Definition	A horizontal axis wind turbine is the one whose axis of rotation is horizontal.	A wind turbine is called a vertical axis wind turbine if its axis of rotation is vertical.
Axis of rotation with respect to wind stream	The axis of rotation of turbine is parallel to the wind stream.	The axis of rotation of the turbine is perpendicular to the wind stream.
Location of gear box and electric generator	The gear box & electric generator is installed at the top of the tower.	The gear box & generator is installed on the ground.
Need of yaw mechanism	The yaw mechanism is required to orient the turbine in the direction of wind.	It does not require yaw mechanism because it receives wind from all directions.
Self-starting	Horizontal axis wind turbine is self-starting.	Vertical axis wind turbine is not self-starting, hence a starting mechanism is required to start it from stationary position.
Design and installation	The design and installation of a horizontal axis wind turbine is complex.	The design and installation of a vertical axis wind turbine is comparatively simple.
Operation space of blades	It requires large space for blade's operation.	It requires small space for blade's operation.
Dependency on wind direction	The operation of horizontal axis wind turbine is dependent on wind direction.	The operation of vertical axis wind turbine is independent of the wind direction because it receives wind from all directions.
Height from ground	The height of the horizontal axis wind turbine from ground is large.	The vertical axis wind turbine is installed at comparatively smaller distance from the ground.
Need of nacelle	In case of horizontal axis wind turbines, a heavy nacelle is installed at the top of the tower.	There is no need of nacelle in case of vertical axis wind turbines.
Power coefficient	It has a high power coefficient.	It has a low power coefficient.
Noise produced	It is noisy.	It is less noise.
Efficiency	The ideal efficiency is around 50% to 60 %.	The ideal efficiency is usually more than 70%.
Hindrance for birds	It cause high obstruction for birds.	It cause less hindrance for birds.
Cost	It is more expensive due their complex design and installation.	It is less expensive because their design and installation is quite simple.

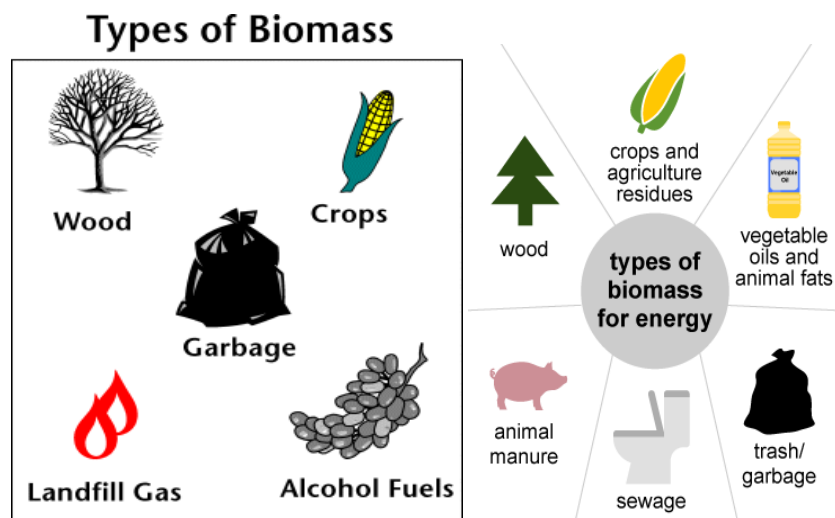


Biomass Energy

- Biomass is renewable organic material that comes from plants and animals. Biomass was the largest source of total annual U.S. energy consumption until the mid-1800s
- **Biomass energy is energy generated or produced by living or once-living organisms.**
- Biomass continues to be an important fuel in many countries, especially for cooking and heating in developing countries.
- The use of biomass fuels for transportation and for electricity generation is increasing in many developed countries as a means of avoiding carbon dioxide emissions from fossil fuel use
- Biomass contains stored chemical energy from the sun. Plants produce biomass through photosynthesis. Biomass can be burned directly for heat or converted to renewable liquid and gaseous fuels through various processes.

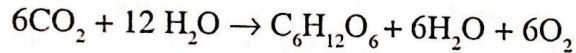
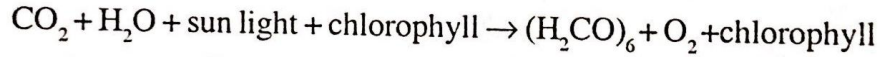
Biomass sources for energy include

- **Wood and wood processing wastes**—firewood, wood pellets, and wood chips, lumber and furniture mill sawdust and waste, and black liquor from pulp and paper mills
- **Agricultural crops and waste materials**—corn, soybeans, sugar cane, switchgrass, woody plants, and algae, and crop and food processing residues, mostly to produce biofuels
- **Biogenic materials in municipal solid waste**—paper, cotton, and wool products, and food, yard, and wood wastes
- **Animal manure and human sewage** for producing biogas/renewable natural gas

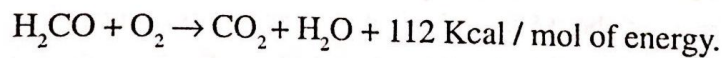


9.2.1 Photosynthetic oxygen production

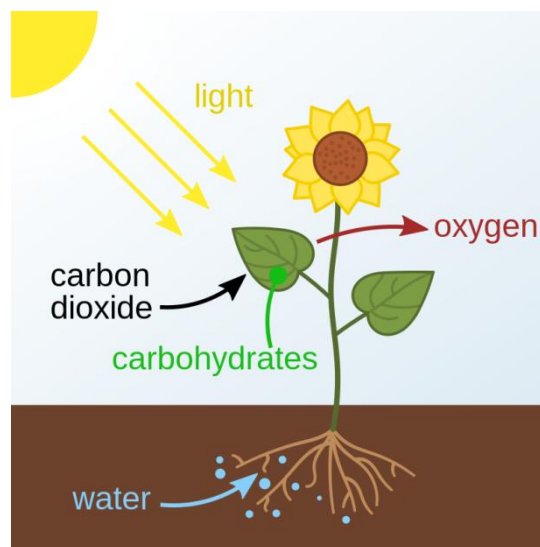
Photosynthesis is a complex process in which water and CO₂ molecules are broken down in sun light and releases carbohydrate and pure oxygen.



The absorbed light is in the ultraviolet and infrared range. The chlorophyll absorbs visible light and passes its energy on to the water molecules and releases an hydrogen atom. The hydrogen atom thus produced reacts with CO₂ molecule to produce H₂CO and O₂. At high temperature H₂CO breaks to release energy.



By growing algae in plastic tubes or ponds, we can produce large amount of carbohydrates. The heat could be generated by burning algae and then could be used to generate electricity by conventional methods.



Converting biomass to energy

Biomass is converted to energy through various processes, including:

- Direct combustion (burning) to produce heat- **Heating the water**
- Thermochemical conversion to produce solid, gaseous, and liquid fuels- **Pyrolysis and Gasification.**
- Chemical conversion to produce liquid fuels- **Transesterification**
- Biological conversion to produce liquid and gaseous fuels- **Anaerobic digestion**

Biogas

Anaerobic fermentation of cellulose containing organic material results in production of a combustible gas which is known as biogas.

Methane forming bacteria then converts the fatty acids into methane, carbon dioxide and water.

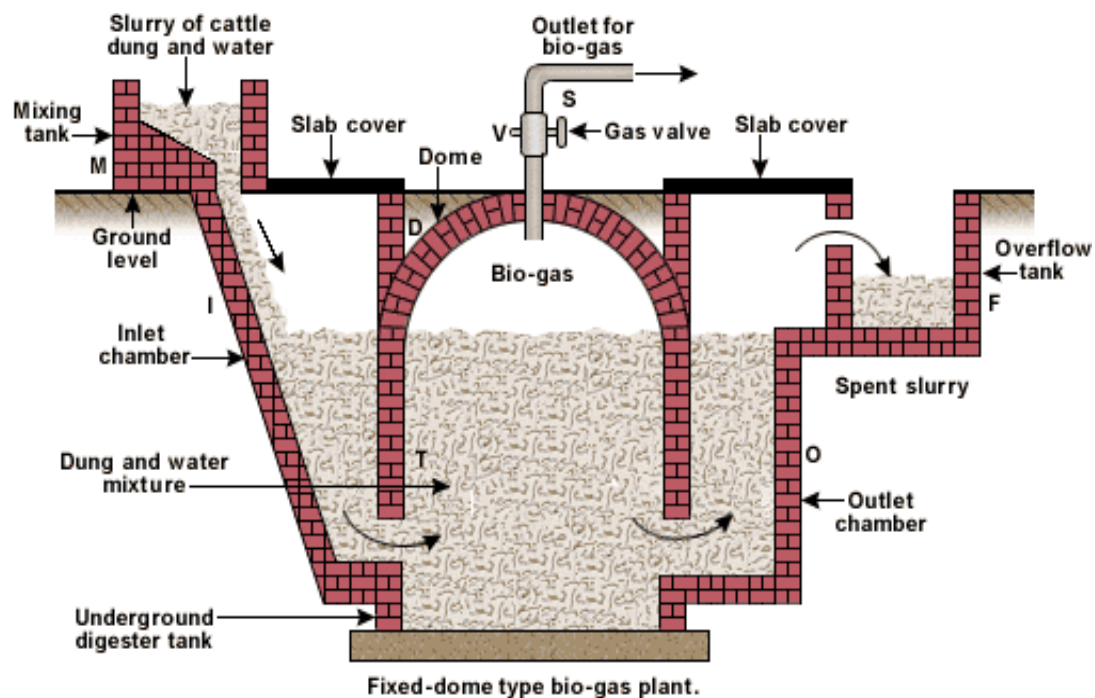
Constituents of biogas

- Methane (50-65%)
- Carbon dioxide (30-40%)
- Hydrogen (1-5%)
- Nitrogen (1%)
- Hydrogen sulphide (0.1%)
- Oxygen (0.1%)
- Water vapours (0.1%)

Biogas plant

The whole system in which **anaerobic fermentation (absence of oxygen)** of cellulose containing organic material takes place and produces biogas, which accumulates in the gasholder or the dome, then utilization of this gas through the gas pipeline under suitable pressure is called biogas plant system.

Fixed dome type of biogas plant





Construction: It consists of a well-like underground tank made of bricks and cement. This tank is called a digester and has inlet and outlet valves. The roof of the tank is dome-shaped. A gas outlet pipe at the top of the dome is fitted. The dome of the digester acts as a storage tank of biogas. There is a mixing tank made above the ground level which is connected to the inlet valve of the digester through a slopping inlet chamber below the ground level. On the other side of the digester, a rectangular tank called an outlet chamber is constructed with bricks and cement. This outlet chamber is connected to the overflow tank which collects the used slurry.

Working: Animals-dung is mixed with water (1:1 ratio) to make a slurry in the mixing tank. This slurry enters the digester through the inlet chamber. The digester is filled partially with slurry so that enough space is left above it in the dome for the collection of biogas. The slurry in the digester is left for about two months for fermentation. Anaerobic microorganisms are responsible for this action. As a result of fermentation, biogas is formed which is collected in the dome. When a sufficient amount of biogas is collected in the dome, it exerts a large pressure on the slurry and forced it to go into the overflow tank through the outlet chamber. The biogas is taken out from the dome through a pipe and used for cooking food or heating water whenever required. Once the biogas plant starts functioning, more and more slurry may be fed into the digester to get the continuous supply of biogas. The used slurry collected in the overflow tank is rich in nitrogen and phosphorus which are essential for the growth of crops and plants. Hence this used slurry can be used as manure.





Advantages:

Fixed dome biogas units have low construction costs, no moving parts, and no rusting steel parts. Hence long life (20 years or more), underground construction, affording protection from winter cold and saving space creates employment locally.

Disadvantages:

Plants often do not gaslight (porosity and crack; gas pressure fluctuates substantially and is usually very high, low digester temperatures.

These plants can only be recommended where experienced biogas technicians can supervise construction.

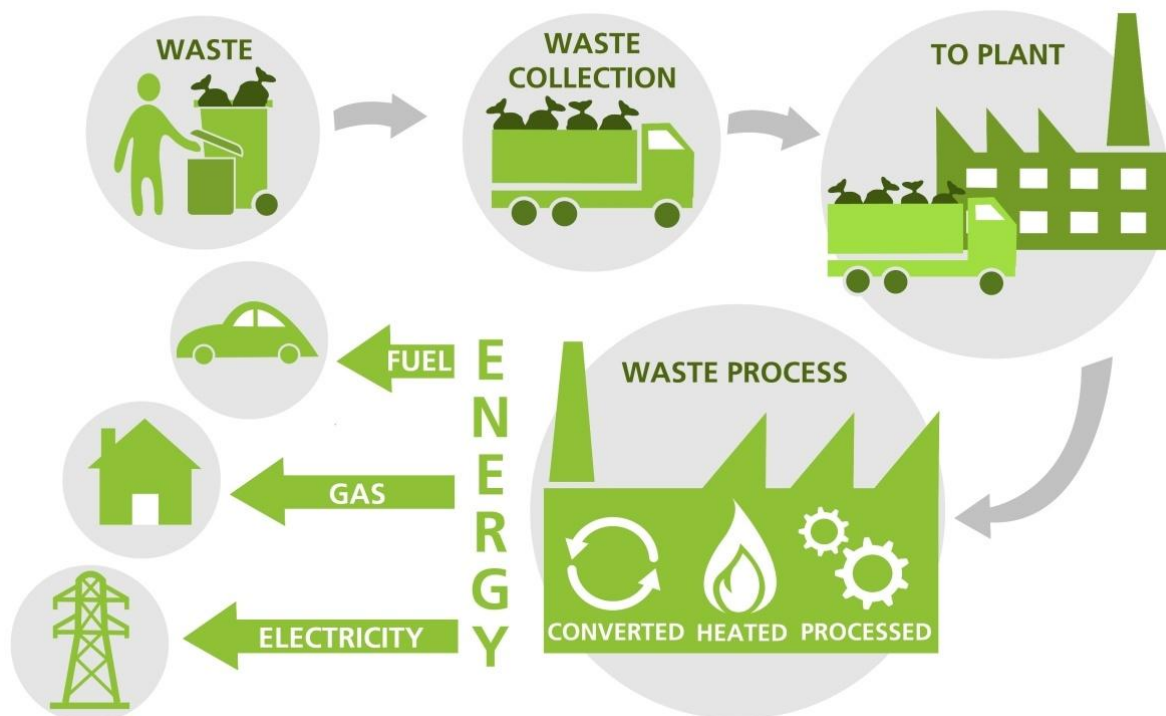
Classification of biogas plants

The biogas plants are mainly classified into three classes

1. Community biogas plants
2. Institutional biogas plants
3. Family size biogas plants

Waste: Unwanted or unusable materials are called waste. The waste is further classified into organic waste (wet), recyclable waste (dry) and reject/sanitary waste.

The following are the steps to convert waste into energy

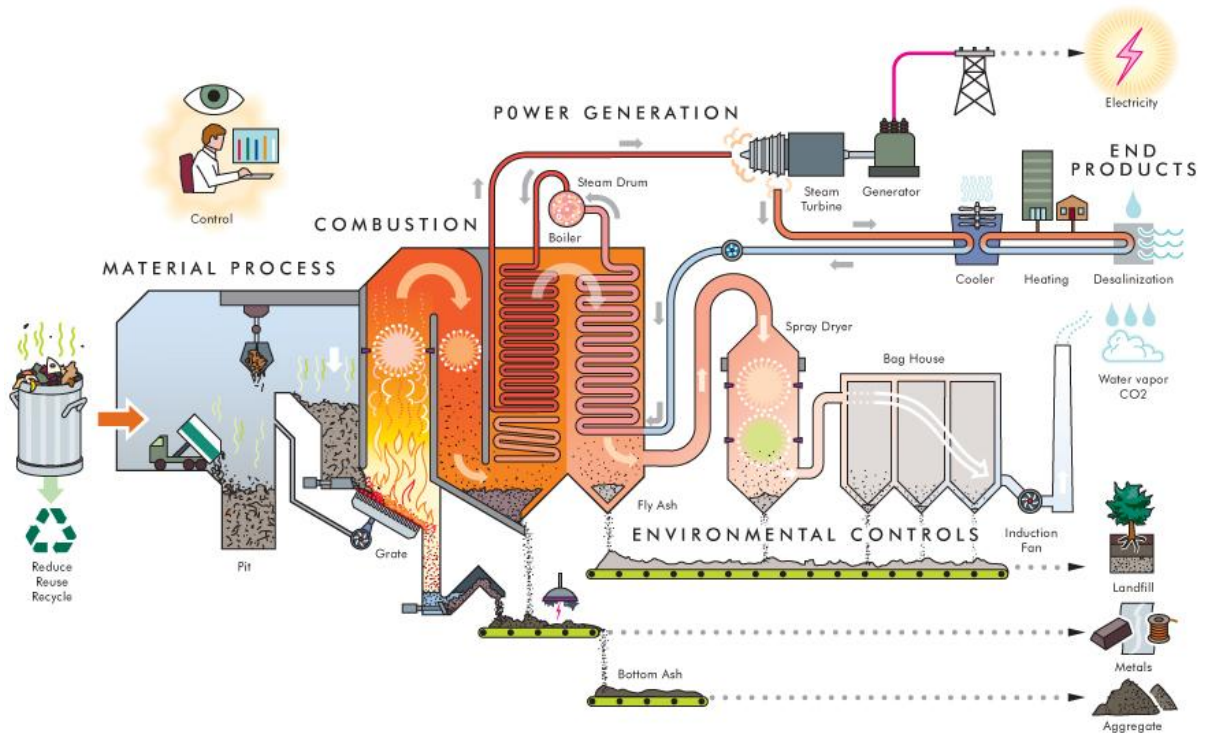


WASTE-TO-ENERGY PLANT

Energy recovery from waste is the conversion of non-recyclable waste materials into usable heat, electricity, or fuel through a variety of processes, including combustion, gasification, Pyrolyzation, anaerobic digestion and landfill gas recovery. This process is often called waste to energy.

1. Combustion
2. Gasification
3. Anaerobic digester
4. Landfill gas

Combustion Waste- To-energy Plant



Waste material is received in an enclosed receiving area, where it is thoroughly mixed in preparation for combustion. Mixed waste enters the combustion chamber on a timed moving grate, which turns it over repeatedly to keep it exposed and burning. Fine airborne particulates (fly ash) are removed in the filter baghouse. The acidic combustion gasses are neutralised with an injection of lime or sodium hydroxide.



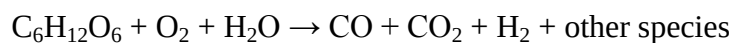
The unburned remains of combustion —"bottom ash"— are passed by magnets and eddy current separators to remove both ferrous (steel and iron) and other metals, such as copper, brass, nickel, and aluminum, for recycling. The remaining ash can be used as aggregate for roadbeds and rail embankments. Superheated steam powers the steam turbine generator. The cooling steam is cycled back into water through the condensor or diverted as a heat source for buildings or industry. Cooled steam is reheated in the economiser and superheated to complete the steam cycle.

Activated carbon (charcoal treated with oxygen to increase its porosity) is injected into the hot gases to absorb and remove heavy metals, such as mercury and cadmium. Nitrogen oxide in the rising burn gases is neutralised by the injection of ammonia or urea. Dioxins and furans are destroyed by exposing flue gases to very high temperatures.

Gasification

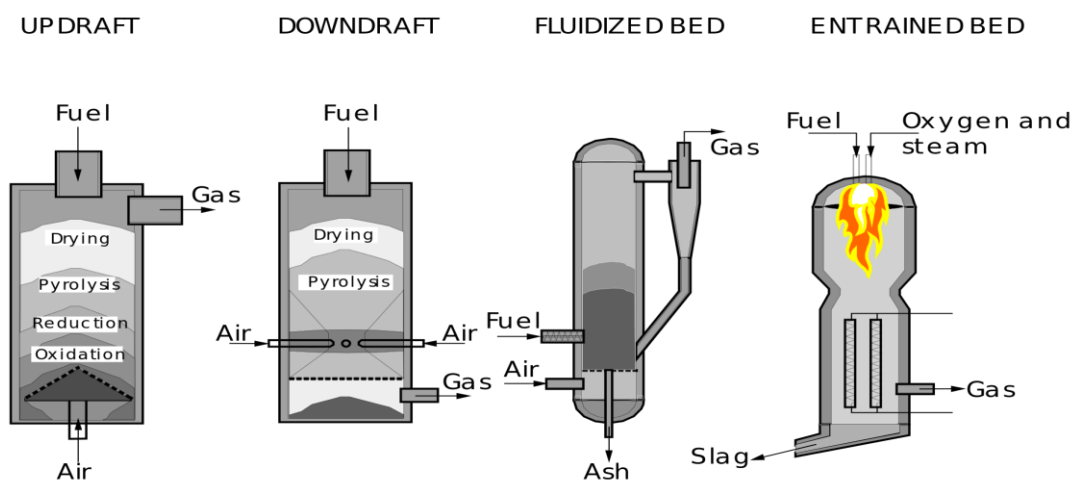
Gasification is a process that converts organic or fossil-based carbonaceous materials at high temperatures ($>700^{\circ}\text{C}$), without combustion, with a controlled amount of oxygen and/or steam into carbon monoxide, hydrogen, and carbon dioxide. The carbon monoxide then reacts with water to form carbon dioxide and more hydrogen via a water-gas shift reaction. Adsorbers or special membranes can separate the hydrogen from this gas stream.

Simplified example reaction

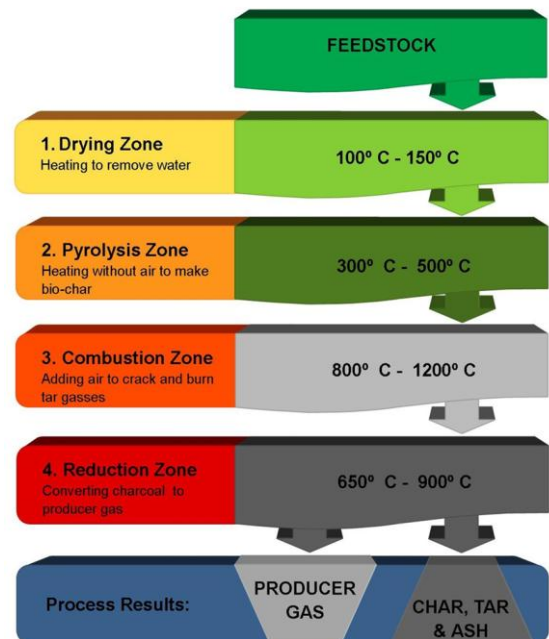
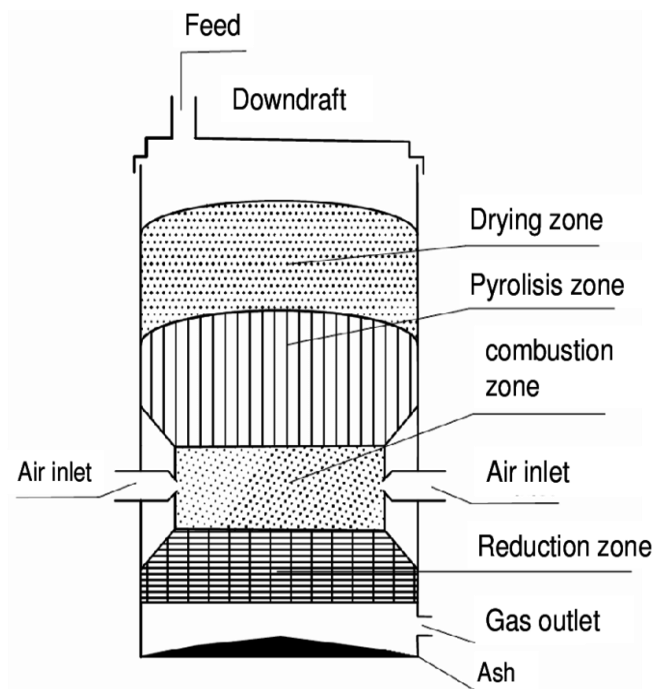


The resulting gas mixture is called syngas (from synthesis gas) or producer gas.

Types of Gasifier



Downdraft Gasifier



In a downdraft gasifier, air or oxygen generally enters the middle zone of the downdraft gasifier above the grate, as presented in above Figure. Air or oxygen enters at or above the oxygen region level in the downdraft gasifiers. The feed materials are entered at the top of the gasifier, similar to the updraft gasifier. However, air and generated gas mixtures are passed through the oxidation region. In a downdraft gasifier, the producer gases are removed at the bottom level of the gasifier. Therefore, gases and fuels in co-current or downdraft gasifiers are moved in the same direction. When the gases and fuels move down, the fuel must pass through a charcoal bed and generate H_2 , CO , CO_2 , and CH_4 . In a downdraft gasifier, based on the hot region temperature and residence time of tars, most of the tar are broken down. Therefore, the generated product gas in co-current or downdraft gasifier contains lower tar than updraft gasifier. Consequently, they are suitable to use in an internal combustion engine compared to the updraft gasifier gases.

*****The End*****